

Pretreatment of sugarcane bagasse with dilute citric acid and enzymatic hydrolysis: Use of black liquor and solid fraction for biogas production

Michelle Garcia Gomes^a, Aline Gomes de Oliveira Paranhos^b, Adonai Bruneli Camargos^b, Bruno Eduardo Lobo Baêta^b, Milla Alves Baffi^c, Leandro Vinícius Alves Gurgel^b, Daniel Pasquini^{a,*}

^a Federal University of Uberlândia, Chemistry Institute, Av. João Naves de Ávila, 2121, Bloco 1D, Campus Santa Mônica, Uberlândia, 38400-902, Minas Gerais, Brazil

^b Federal University of Ouro Preto, Institute of Exact and Biological Sciences, Campus Universitário Morro do Cruzeiro, s/n°, Bauxita, Ouro Preto, 35400-000, Minas Gerais, Brazil

^c Federal University of Uberlândia, Agricultural Sciences Institute, Av. Amazonas, s/n°, Bloco 2E, Campus Umuarama, Uberlândia, 38405-320, Minas Gerais, Brazil

ARTICLE INFO

Article history:

Received 31 August 2021

Received in revised form

24 March 2022

Accepted 10 April 2022

Available online 14 April 2022

Keywords:

Biorefinery

Biomass pretreatment

Enzymatic hydrolysis

Anaerobic digestion

Biogas

ABSTRACT

This study evaluated the pretreatment of sugarcane bagasse (SCB) with dilute citric acid solution for biogas production. The black liquor (BL) from pretreatment and residual solid fraction (RSF) obtained after enzymatic hydrolysis were used as substrates for the biogas production. After saccharifications of the pretreated solid fractions performed with Cellic® CTec 3 cocktail, the highest concentration of total reducing sugars (TRS) was in the range of 10.7–184.8 g L⁻¹ and the sugar yields were in the range of 3.5–88.8%. For BL, the contents of glucose, xylose, and arabinose were in the range of 1.9–8.9 g L⁻¹, 4.0–24.7 g L⁻¹, and 0.2–5.4 g L⁻¹, respectively. Biogas production from BL and RSF resulted, respectively, in maximum values of 563.6 and 57.8 NmL g_{CO₂}⁻¹, indicating the feasibility of using dilute citric acid pretreatment coupled to biogas production in biorefineries. Brief energy and economic assessment were carried out considering the energy balance (thermal energy generated by burning biogas - thermal energy spent in the pretreatment), as well as revenue from a possible sale of electricity generated in a combined heat and power system.

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1. Introduction

Lignocelluloses are one of the most abundant and renewable biomasses in the world [1]. They are generally the most widely available low-cost renewable resources to consider in the production of biocompounds and chemicals [2]. This class of compounds includes lignocellulosic residues, such as sugarcane bagasse (SCB). Lignocelluloses have a complex molecular structure, with tangled chains of cellulose, hemicelluloses, and lignin, which are very slowly degraded by microorganisms [3]. In a lignocellulosic biorefinery, the use of these residues for the production of various by-

products invariably require a pretreatment step to improve bioconversion, i.e., the access of hydrolytic enzymes to their constituents in the enzymatic hydrolysis step [4].

The interlaced structure, in addition to lignin, provides a protective barrier to cellulose chains and affects their enzymatic digestibility [5,6]. Consequently, it leads to lower yields of fermentable sugars [3]. Acid pretreatment can partially dissolve hemicelluloses from lignocellulosic biomasses because the glycosidic bonds in hemicelluloses are susceptible to acid hydrolysis. Furthermore, it can reduce the amount of lignin present in pretreated biomasses, since alkyl-aryl-ether bonds are also susceptible to acid hydrolysis [3]. Therefore, the pretreatment step can improve the subsequent production of value-added molecules and/or other biocompounds, such as biogas, bioethanol, among others [4]. However, pretreatment methods are challenging and complex, but they are considered a fundamental step to improve the efficiency of

* Corresponding author.

E-mail addresses: daniel.pasquini@ufu.br, danielpasquini2005@yahoo.com.br (D. Pasquini).

subsequent processes [5]. The main challenges faced during the conversion of lignocellulosics into biofuels are associated with the complex structure of biomasses and their components, such as the barrier of hemicelluloses and lignin that restrict the action of cellulases, the crystalline structure of cellulose, the small accessible surface area, and the high cost of enzyme cocktails used to hydrolyze polysaccharides [7].

During pretreatment, the compact lignocellulosic structure is fragmented, and the cellulose fiber is exposed, becoming more accessible to enzymes [7], with partial or total solubilization of hemicelluloses and partial solubilization of lignin [5]. In addition, pretreatments can be optimized to reduce the lignin and/or hemicelluloses content, decrease the cellulose crystallinity, reduce the particle size, increase the surface area and porosity of the biomass [8]. In addition, the choice of pretreatment must also consider the minimization of the production of substances that inhibit enzymes [9] and the metabolism of microorganisms [10].

Generally, chemical pretreatments involving concentrate and dilute inorganic acids are efficient in the hydrolysis of lignocellulosics [4], but lead to low sugar yields due to degradation of monomeric sugars into furans and organic acids, producing undesirable by-products that are inhibitors of the fermentation step [11]. A strategy to reduce the degradation of monosaccharides and the formation of fermentation inhibitors at elevated temperatures is to carry out a dilute acid pretreatment in two or more steps [11]. This way, hemicelluloses are hydrolyzed under milder conditions and cellulose is hydrolyzed under more severe conditions in separated steps with the removal of the liquid fraction in each step, which increases the sugar yield and decreases the formation of inhibitors [11]. In addition, the use of strong acids causes corrosion of reactors and, in addition, the cost of acids is relatively high. Then, they must be recovered to make the process more viable [12].

Inhibitors commonly generated during acid pretreatments include acetic acid [12], released by the deacetylation of hemicelluloses, 2-furfuraldehyde (FF) and 5-hydroxymethyl-2-furfuraldehyde (HMF), which are formed by the degradation of pentoses (xylose and arabinose) and hexoses (glucose, mannose, and galactose), respectively [13]. In addition, FF and HMF can be degraded to formic acid and levulinic acid [14].

Alternatively, some dilute acids have been widely recognized and used due to their numerous advantages, such as low cost, availability, and efficiency in altering the lignocellulosic structure and at the same time leading to low inhibitor formation [15]. It has been reported that dilute acid pretreatments can improve the efficiency of the enzymatic hydrolysis step, mainly for grassy biomass, increasing the accessibility to cellulose and altering lignin distribution [3,16]. For acid-pretreated bamboo and woody biomass, the enzymatic digestibility is still lower compared to grassy biomass. According to Huang et al. [16], an alternative to increase the enzymatic digestibility of acid-pretreated bamboo is to couple the acid pretreatment with the post-extraction process using phosphoric acid, urea, or ethanol.

The use of dilute citric acid in the pretreatment of biomass is attractive, since it is less corrosive to the equipment and is a low-cost and environmentally friendly compound [17]. In addition, it has low toxicity [18] and is easily biodegradable [19], reducing costs [20] and making the pretreatment step more efficient. However, there are no previous studies reporting the pretreatment of sugarcane bagasse with dilute citric acid and the use of the pretreated fraction (solid and liquid phases) as substrates in anaerobic digestion (AD) for biogas production. After pretreatment, the cellulose-rich solid fraction is submitted to the enzymatic hydrolysis step and a series of cellulolytic and hemicellulolytic enzymes act in synergism, producing glucose and xylose monomers, respectively [21]. Xylose is the main hemicellulose sugar and can be used as a

carbon and energy source in fermentation processes [22] after a detoxification step to remove fermentation inhibitors [23]. Typically, the main purpose of enzymatic hydrolysis step is the production of fermentable sugars to produce second-generation (2G) ethanol or cellulosic ethanol and bioproducts.

Alternatively, waste generated during biomass processing in a lignocellulosic biorefinery can be reused in AD to produce biogas. Biogas is basically composed of methane (CH_4), hydrogen (H_2), carbon dioxide (CO_2), and hydrogen sulfide (H_2S), with different compositions and concentrations. This variation may be due to the characteristics of substrates, the type of biodigesters, and the operational parameters (e.g. applied organic load, agitation (batch) or inlet flow rate (continuous), temperature, and food-to-inoculum (F/I) ratio) [24]. Biogas is considered a clean and an environmentally friendly biofuel and has been used to generate heat, steam, and electricity, as well as an automotive fuel [25].

AD of by-products generated from biomass processing for biogas production requires less capital investment per unit of production, compared to other renewable energies such as hydroelectric, solar, and wind power [26]. This favors the implementation of AD units in first-generation (1G) biorefineries that produce first-generation (1G) ethanol and sugar from sugarcane juice, to produce biogas using biorefinery waste such as filter cake, vinasse, fly ash, among others as substrates for AD. These 1G biorefineries can also be integrated with 2G biorefineries to produce 2G ethanol using surplus sugarcane bagasse from the 1G ethanol and sugar production process. This way, another alternative substrate to produce biogas is the black liquor (BL) generated after the pretreatment step [27].

A detailed techno-economic analysis of this type of integration was not investigated yet. However, in the study by Patinvoh et al. [10], the cost of producing methane from wheat straw using steam pretreatment was estimated at USD 0.85 m^{-3} , with the pretreatment cost at USD 0.11 m^{-3} CH_4 (13% of the total cost). The 2G ethanol production from lignocellulose residue in California was estimated at USD 0.43 L^{-1} ethanol and could be reduced on average by USD 0.10 L^{-1} if the pretreatment cost is paid for biogas production. Uellendahl et al. [28] demonstrated that the net energy gain is greater as the specific yield of methane increases significantly when applying a wet oxidation pretreatment to raw perennial crops when compared to non-pretreated annual crops such as corn. When applying a pretreatment step to perennial crops, the output-to-input energy ratio and cost-to-benefit ratio for the entire biomass supply chain and conversion into biogas becomes higher than for non-pretreated corn. Therefore, a pretreatment step makes the use of perennial crops more competitive than the use of non-pretreated corn. In addition, the combination of perennial crops with annual crops makes biogas production from energy crops more sustainable.

Hu et al. [29] mentioned in a technical report by the National Renewable Energy Laboratory (NREL) and Harris Group Inc., that the pretreatment accounted for 14.2% of capital costs and >15.9% of manufacturing costs and the enzyme cocktail accounted for 21.3% of capital costs. This shows that the pretreatment step is always essential to make polysaccharide fraction available for subsequent steps, i.e., enzymatic hydrolysis and fermentation to ethanol.

Regarding the biorefinery concept, this study evaluated the pretreatment of raw SCB in a pressurized system using dilute citric acid solution, with temperature (T), time (t), and percentage by weight of citric acid (PCA) in relation to raw SCB as the operational variables, aiming the application of pretreated solid and liquid fractions in enzymatic hydrolysis and biogas production, respectively. It is worth mentioning that the first part of this study on dilute citric acid pretreatment and its effects on enzymatic hydrolysis was previously published by our research group. In this first

study, dilute citric acid pretreatment was carried out under different conditions to increase the accessibility to cellulose fraction and, consequently, improve the enzymatic hydrolysis yield [30]. Glucose concentration reached a maximum of 23.74 g L^{-1} for SCB samples pretreated at temperatures of $160\text{--}180^\circ\text{C}$, $5.25\text{--}9.75 \text{ wt\%}$ of PCA, and reaction times of $55\text{--}90 \text{ min}$, while xylose reached a maximum of 5.23 g L^{-1} for SCB samples pretreated at a temperature of 150°C , $7.5\text{--}12 \text{ wt\%}$ of PCA, and 55 min [30]. Therefore, the present study, which is a continuation of the previous study, presents the results of biogas production from the liquid fraction (black liquor) obtained after dilute citric acid pretreatment of SCB and from the residual solid fraction (RSF) generated after enzymatic hydrolysis of pretreated SCB. The pretreatment condition employed for biogas production was the one of the best pretreatment conditions studied, with the greatest release of fermentable sugars. Concomitantly, a brief economic-energy estimate was made to investigate the feasibility of applying dilute citric acid pretreatment on sugarcane bagasse for production of fermentable sugars and biogas.

2. Materials and methods

2.1. Sugarcane bagasse preparation

Raw SCB was supplied by the Jatiboca Usina de Açúcar e Ethanol (Urucânia, MG, Brazil). Raw SCB was washed with distilled water at 60°C for 1 h . Then, the wet SCB was filtered off to remove excess water and stored at temperatures below 0°C until processing.

2.2. Dilute citric acid pretreatment of sugarcane bagasse

For the dilute citric acid pretreatment of SCB, 30.0 g of raw SCB (dry basis) were added to a 475 mL 316L-grade stainless steel tubular reactor (autoclave type) with 150 mL of aqueous citric acid solution, as described elsewhere [30]. This mixture was heated in a 25 L thermostatic bath (Marconi®, model MA-159) containing glycerin as heating fluid. Pretreatment conditions were investigated using the 2^3 Doehlert experimental design (DED), with five replicates at the center point (CP; samples 6, 7, 9, 10, and 12). The variables investigated were temperature (T), from 120 to 180°C , reaction time (t), from 20 to 90 min , and percentage by weight of citric acid (PCA) to SCB, from 3 to $12 \text{ wt\%}$. The DED matrix is shown in Table 1.

After each pretreatment, the reactor vessel was immediately cooled in an ice-water bath and opened. The solid (SF) and liquid (BL) fractions of the pretreated SCB samples were separated by pressing in a hydraulic press at 9 tons (SOLAB®, model SL-10). The liquid fraction released after pressing was stored in hermetically sealed containers in a freezer. Solid fractions of the pretreated SCB samples were characterized in terms of yield and chemical composition (cellulose, lignin, and hemicelluloses) and used in enzymatic hydrolysis. Enzymatic hydrolysis data were previously reported by our research group [30]. In the present study, the liquid fraction (BL) and solid fraction recovered after enzymatic hydrolysis (RSF) were used for biogas production by anaerobic digestion.

The liquid (BL) and enzymatic hydrolysates) and solid (SF) fractions of pretreated SCB samples obtained from each pretreatment were characterized by high performance liquid chromatography (HPLC) for quantification of sugars (glucose, xylose, and arabinose), organic acids (acetic acid and formic acid) and furans (2-furfuraldehyde (FF) and 5-hydroxymethyl-2-furfuraldehyde (HMF)). The chromatographic system consisted of a Shimadzu® HPLC system equipped with Aminex HPX 87H ($300 \times 7.8 \text{ mm}$, BioRad®) column maintained at 55°C , using a refractive index detector (Shimadzu®, model RID- 6A) for sugars and a UV-VIS detector (Shimadzu®, model SPD-10AV) set at the wavelengths of

Table 1

Experimental design matrix (DED) used in the dilute citric acid pretreatment of sugarcane bagasse.^a

Experimental condition	PCA (wt.%)	T (°C)	t (min)
1	5.25	120	55
2	9.75	120	55
3	7.5	130	90
4	5.25	140	20
5	9.75	140	20
6	7.5	150	55
7	7.5	150	55
8	12.0	150	55
9	7.5	150	55
10	7.5	150	55
11	3.0	150	55
12	7.5	150	55
13	9.75	160	90
14	5.25	160	90
15	7.5	170	20
16	9.75	180	55
17	5.25	180	55

^a Data previously published by our research group [30].

210 nm and 274 nm for organic acids and furans, respectively. Sulfuric acid (5 mmol L^{-1}) was used as eluent at a flow rate of 0.6 mL min^{-1} [27].

2.3. Enzymatic hydrolysis assays

The operational parameters (temperature of $45\text{--}50^\circ\text{C}$ and pH of $5.0\text{--}5.5$) of the enzymatic hydrolysis tests were defined according to the recommendations of the Cellic® CTec 3 supplier [31]. The activities of Cellic® CTec 3 HS cocktail include advanced AA9 molecules, *T. reesei* cellulases, heterogeneously expressed β -glucosidases, and hemicellulases. The FPase activity (Filter paper activity) of this enzyme complex was 264 FPU g^{-1} , according to Ghose [32], and it contains a minimal of 1.700 BHU (Biomass Hydrolysis Units - BHU(2)-HS) per gram of product, measured by fluorescence enhancement marker. The liquid-to-solid ratio used in the tests was 10 mL g^{-1} , with 1.0 g of solid fraction (dry basis) for each test. The initial enzyme concentration was fixed at 10 FPU g^{-1} [30] in relation to the cellulose content present in raw SCB or pretreated SCB samples [32]. The enzyme complex was added to reaction flasks, containing raw SCB or pretreated SCB samples with 10 mL of sodium citrate buffer (pH 4.8), under stirring of 150 rpm at 50°C . The assays were performed in duplicate. After 72 h , the suspensions were filtered and the hydrolysates were centrifuged (NOVA® Benchtop Microcentrifuge, model NI1801) for 10 min at 8000 rpm and 4°C , allowing the separation of the liquid and solid fractions [30].

The liquid fractions (hydrolysates) were submitted to HPLC analysis for quantification of sugars (glucose, xylose, and arabinose) under the same chromatographic conditions described in Section 2.2. In addition, a sample of the liquid fraction was collected to determine the concentration of total reducing sugars (TRS) by the 3,5-dinitrosalicylic acid (DNS) method [33]. The RSF obtained after enzymatic hydrolysis was then washed with distilled water until reaching pH ~ 7 and dried in an air circulation oven at 105°C for 12 h . The RSF samples were stored in hermetically sealed zip lock plastic bags in a freezer prior to anaerobic digestion tests. The moisture content of the RSF was determined in triplicate on a moisture analyzer (OHAUS®, model MB23).

2.4. Anaerobic digestion assays

The biochemical biogas potential (BBP) of the BL fraction obtained from dilute citric acid pretreatment of SCB and the RSF

generated after enzymatic hydrolysis of pretreated SCB were evaluated in anaerobic digestion tests. These tests were performed with liquid (BL) and solid (RSF) fractions obtained from the dilute citric acid pretreatment of SCB in condition 2. This condition was chosen due to its lower content of inhibitors, associated with its adequate composition in terms of type and amount of sugars. Other justifications for choosing condition 2 for biogas production will be presented later in the results and discussions section.

The chemical oxygen demand (COD) of the RSF generated after enzymatic hydrolysis of pretreated SCB and the BL generated after pretreatment of SCB were determined according to the closed reflux colorimetric method [34]. In addition, volatile solids (VS) of the anaerobic inoculum and the RSF sample were also determined in accordance with the Standard Methods (methods 2540 B, 2540 E, and 2540 F) [35]. All anaerobic digestion tests were performed under mesophilic conditions in 120 mL glass bottles, which were placed in a thermostatic orbital shaker incubator (35.1 ± 0.3 °C) at 200 rpm. The useful volume of each glass bottle was 60 mL and the remaining volume was headspace.

BL samples (6.0 mL) were added to glass bottles to maintain a F/I ratio of $0.4 \text{ gCOD gVS}^{-1}$ [36]. Each bottle contained varied volumes of inoculum depending on the COD concentration in each sample. A total solids content of approximately 15% and a F/I ratio of 0.4 gVS gVS^{-1} were maintained in tests performed with the RSF sample ($0.89 \text{ gVS g sample}^{-1}$). In each experiment, a solution containing sodium bicarbonate (pH buffer) and micro- and macronutrients (stock solution of 5.0 g L^{-1}) [12] was added according to substrate concentration. To complete the 60 mL of useful volume, distilled water was added and the initial pH in each batch reactor was maintained at 7–8 with NaHCO_3 (1.0 mol L^{-1}) [37]. A control test, without addition of substrate, was also performed to verify the endogenous methanogenic activity of the inoculum.

The anaerobic inoculum used for biogas production was taken from a UASB mesophilic reactor fed with domestic sewage, operated on a demonstration scale at the Sanitation Research and Training Center (CePTS) of UFMG-COPASA, located at the Arrudas WTP Plant (Belo Horizonte, MG, Brazil). The inoculum was pre-incubated for four days (35 °C) to offset biogas production from biodegradation of any residual organic matter. For the RSF sample, the inoculum was concentrated to obtain the minimum water content. The volatile solids (VS) of the anaerobic inoculum were 2.36 g and total suspended solids (TSS) of the sludge were 0.0974 g .

BBP tests were monitored daily by recording bottle pressures using a manometer (Manometer®, model PM-9100HA). All gas production values were normalized considering normal conditions of temperature and pressure (273.15 K and 101.315 Pa), as defined by IUPAC (International Union of Pure and Applied Chemistry) and expressed in NmL (accumulated volume of biogas) and NmL gCOD^{-1} (accumulated volume of biogas per COD added). All experiments were carried out in duplicate.

2.5. Kinetic analysis

The modified Gompertz equation (Eq. (1)) was used to fit the experimental data and estimate the cumulative biogas production. The modified Gompertz model is an empirical model that allows the estimation of important kinetic parameters of anaerobic digestion, such as lag phase time, specific biogas production rate, and maximum cumulative biogas production [38].

$$H = P \exp \left\{ - \exp \left[\frac{R_m e}{P} (\lambda - t) + 1 \right] \right\} \quad (1)$$

where H is the cumulative volume of biogas produced (NmL gCOD^{-1}), P is the potential for biogas production (NmL gCOD^{-1}), R_m is the maximum

biogas production rate (NmL $\text{gCOD}^{-1} \text{ day}^{-1}$), λ is the lag phase time (days), t is the incubation time (days), and e is equal to 2.71828.

2.6. Brief energy and economic assessment

The energy feasibility of the proposed process considering two scenarios was estimated by an energy balance (ΔE) that considered the energy spent ($E^{(-)}$) (Eq. (2)) in the dilute citric acid pretreatment of SCB in condition 2 and the energy obtained ($E^{(+)}$) (Eq. (3)) by burning the methane generated from anaerobic digestion tests of the BL and RSF, in a combined heat and power (CHP) system.

$$E^{(-)} / \text{MJ kg}_{\text{SCB}}^{-1} = C_{p, \text{H}_2\text{O}} w_{\text{H}_2\text{O}} (T_f - T_i) \quad (2)$$

$$E^{(+)} / \text{MJ kg}_{\text{SCB}}^{-1} = \left(\frac{\text{NmL}_{\text{CH}_4}}{w_{\text{COD}}(\text{g})} \right) \left(\frac{w_{\text{COD}}(\text{g})}{w_{\text{BL or RSF}}(\text{g})} \right) \left(\frac{w_{\text{BL or RSF}}(\text{g})}{w_{\text{SCB}}(\text{g})} \right) \left(\frac{1000 \text{ g}_{\text{SCB}}}{\text{kg}_{\text{SCB}}} \right) \left(\frac{10^{-6} \text{ m}^3}{\text{mL}} \right) \left(\frac{34.5 \text{ MJ}}{\text{Nm}^3 \text{CH}_4} \right) \quad (3)$$

$$\Delta E / \text{MJ kg}_{\text{SCB}}^{-1} = E^{(+)} - E^{(-)} \quad (4)$$

where, $C_{p, \text{H}_2\text{O}}$ is the average heat capacity of water ($4.19 \times 10^{-3} \text{ MJ}(\text{kg } ^\circ\text{C})^{-1}$), $w_{\text{H}_2\text{O}}$ is the weight of water used (5 kg of water per 1 kg of raw SCB), T_f is the pretreatment temperature in condition 2 (120 °C), T_i is the initial water temperature (25 °C for the first scenario; 100 °C for the second scenario), NmL_{CH_4} is the final cumulative volume of methane produced, w_{COD} is the weight of chemical oxygen demand (g), w_{SCB} is the weight of SCB (g), w_{BL} is the weight of BL recovered after pretreatment (85 g), w_{RSF} is the weight of RSF recovered after enzymatic hydrolysis of pretreated SCB (22.17 g).

For the calculation of $E^{(-)}$, the first scenario considered water at 25 °C, while the second scenario considered water in the vapor state (100 °C), since it is possible to use cooling water from the CHP system in a heat exchanger to heat and vaporize the feed water used in the pretreatment. The calculation of $E^{(-)}$ after pretreatment was simplified and took into account only the energy spent to heat the water to the pretreatment temperature (120 °C), since the reaction vessel operated with 20% solids in this condition.

For the calculation of $E^{(+)}$, the enthalpy of combustion (ΔH°) of methane of $34.5 \text{ MJ}(\text{Nm}^3 \text{CH}_4)^{-1}$ was considered, as well as the pretreatment yield reported in this study and the conversion efficiencies of the CHP system described by Cano et al. [39]. According to Cano et al. [39], a conventional CHP system has a combustion efficiency of 85%, with the conversion of chemical energy into thermal energy corresponding to 65% and the conversion to electrical energy corresponding to 35%. The energy balance (ΔE) in the proposed process is given by the difference between $E^{(+)}$ and $E^{(-)}$.

The economic assessment was carried out considering an electricity market value (CHP system and pretreatment cost) of $0.08 \text{ USD kWh}^{-1}$, according to the Brazilian Energy Research Company [40]. In addition, all the calculations presented in this study were converted to the calculation basis per ton of raw SCB.

3. Results and discussion

3.1. Dilute citric acid pretreatment of sugarcane bagasse

Dilute citric acid pretreatment positively affected the structure of pretreated samples, promoting chemical and compositional

changes in the lignin, hemicelluloses, and cellulose fractions in comparison to raw SCB. These changes resulted in higher enzymatic digestibility of cellulose fraction, as reported by Gomes et al. [30]. The chemical composition of the pretreated SCB samples was in the range of values reported in literature for moisture content (2.11–2.58%), cellulose (41–52%), hemicelluloses (23–36%), and lignin (<20%). A decrease in lignin and hemicelluloses content was observed after pretreatment, with cellulose preservation occurring preferentially in lower levels of PCA and *t* and higher levels of *T*. The compositional analysis for the BL generated after pretreatment of SCB with dilute citric acid under different conditions is shown in Table 2.

Figs. 1 and 2 show the response surface graphs for the contents of components present in BL, as a function of the variables studied (*T*, *t*, and PCA). For inhibitors (formic acid, acetic acid, FF, and HMF), it was observed a direct relationship between the content of these compounds and the experimental conditions investigated, with a similar behavior for all of them (Fig. 1). It suggests that the increase in *T*, *t*, and PCA can lead to an increase in the generation of inhibitors [41]. Therefore, to avoid the use of these black liquors for other processes rather than those employing enzymes or microorganisms, it would be more suitable to work with pretreated samples under milder *T*, *t*, and PCA conditions. The low concentration of formic acid and the absence of levulinic acid in the black liquors may indicate that the degradation of furans was moderate. Higher generation of formic acid would be expected under more severe pretreatment conditions [42]. According to Fangkum and Reungsang [12] and Aguilar et al. [22], acetic acid can be an inhibitor of microbial growth when present in concentrations ranging from 4 to 10 g L⁻¹. It can cross cell membranes and lower intracellular pH, thus affecting the metabolism of the microorganisms. In the present study, acetic acid was detected, and the maximum concentrations observed were 9.54, 6.45, and 5.36 g L⁻¹ in samples 16, 13, and 14, respectively (Table 2). For other samples, the minimum concentrations observed were below the toxic level. Acetic acid and formic acid were the most abundant inhibitory compounds found in black liquors obtained under different pretreatment conditions investigated in this study. The highest concentrations of acetic acid and formic acid were found in black liquors obtained under pretreatment conditions in which hemicelluloses were removed to a greater extent. In contrast, FF and HMF concentrations were relatively low (Table 2). According to Fangkum and Reungsang [12] and Fockink et al. [14], formic acid has a more inhibitory effect than

acetic acid due to its smaller molecular size, which facilitates its diffusion through the membranes of microorganisms, leading to greater toxicity.

The response surface graphs generated for the contents of glucose, xylose, and arabinose in black liquors are shown in Fig. 2. For glucose, the highest contents were obtained under conditions of high values of *T*, *t*, and PCA, i.e., a behavior similar to that observed for the formation of inhibitors. On the other hand, for xylose and arabinose, it was observed that the highest contents of these compounds were obtained at longer reaction times and higher PCA contents, but at lower temperatures. This result indicated that hemicelluloses were degraded under milder reaction conditions, and that with increasing reaction temperature, a greater attack was observed in the cellulose fraction, which resulted in the release of glucose in the black liquor. It can also be inferred that, at higher temperatures, there was a favorable conversion of sugars into inhibitors. As observed, less severe pretreatment conditions led to the production of black liquors with higher contents of arabinose (samples 2 and 3). Arabinose is a compound that is generally bound in the hemicellulose chains in the form of branches [12], and therefore, it is more easily hydrolyzed compared to xylose, which is the main constituent of hemicelluloses in SCB [13]. In addition, arabinose is degraded to FF faster than xylose, since the activation energy to convert arabinose to FF (103.8 kJ mol⁻¹) is lower than that of xylose to FF (119.3 kJ mol⁻¹) [43,44]. Therefore, samples with higher arabinose contents are samples with lower C5 sugar degradation, and consequently, they can be strong candidates to be applied for biogas production tests, as presented in the present study. As expected, these samples (samples 2 and 3) also showed lower contents of inhibitory compounds.

3.2. Enzymatic hydrolysis

Table 3 shows the chemical composition of the hydrolysates obtained from the enzymatic hydrolysis of samples of raw SCB and pretreated SCB. Glucose and xylose data presented in Table 3 were previously published by our research group [30]. TRS and arabinose values are shown in the present study. The lower yields, i.e., lower residual weights after enzymatic hydrolysis (Table 3), observed in samples 17, 16, and 14, can be attributed to the severity of the pretreatments, with the temperatures higher than 160 °C, reaction times longer than 55 min, and PCA contents of 5.25 and 9.75 wt%, respectively.

In our previous study, it was observed that the most significant positive effect on responses was related to the variable *T*, which has been confirmed by the highest concentrations of glucose and xylose obtained in pretreatments carried out at higher temperatures [30]. It was possible to observe that the other variables studied had a smaller effect on enzymatic conversion. Glucose concentration reached a maximum of 23.74 g L⁻¹ for the sample pretreated at 160 °C and 90 min with PCA of 5.25 wt% (sample 14), while xylose concentration reached a maximum of 5.23 g L⁻¹ in the sample pretreated at 150 °C and 55 min with PCA of 7.5 wt% (sample 9). In addition, the digested pretreated solid fraction after enzymatic hydrolysis (100% - Yield; Table 3) reached a maximum of 88.8%, with glucose and xylose concentrations of 23.7 and 0.23 g L⁻¹, respectively, for the sample pretreated at 180 °C and 55 min with PCA of 5.25 wt% (sample 17). This condition led to a very similar glucose concentration compared to sample 14, but with a lower xylose concentration, showing that an increase in the pretreatment temperature from 150 to 180 °C led to a decrease in the xylan content in pretreated SCB. This result is expected as the activation energy for xylan hydrolysis into xylose (105.0 kJ mol⁻¹) is lower than the activation energy for glucan hydrolysis into glucose

Table 2
Composition of the black liquors obtained after different pretreatment conditions of sugarcane bagasse with dilute citric acid.

Sample	Compounds present in black liquor (g L ⁻¹)						
	Glucose	Xylose	Arabinose	Formic Acid	Acetic Acid	HMF	FF
1	6.20	20.22	3.34	0.07	0.07	0.01	0.05
2	5.50	20.44	5.38	3.34	2.25	0.15	0.03
3	4.85	17.22	4.12	2.36	0.49	0.03	0.10
4	2.86	13.36	0.19	2.43	0.03	0.00	0.05
5	4.61	16.07	3.14	2.41	0.23	0.03	0.04
6	4.34	18.72	3.84	4.48	2.57	0.54	0.15
7	4.00	16.29	3.53	4.36	2.11	0.52	0.12
8	4.53	17.86	2.51	3.57	2.72	0.43	0.19
9	4.05	15.63	3.81	3.54	2.10	0.47	0.10
10	1.90	7.15	1.20	1.49	0.85	0.16	0.04
11	4.04	13.82	3.28	3.81	1.57	0.25	0.07
12	4.97	21.91	3.24	4.29	2.98	0.47	0.13
13	8.57	18.78	2.47	6.31	6.45	1.39	1.61
14	8.85	24.70	3.22	7.66	5.36	1.51	0.66
15	4.89	17.06	2.87	3.20	1.30	0.26	0.08
16	9.15	17.03	1.28	7.89	9.54	1.79	3.12
17	2.43	4.03	0.28	1.49	0.95	0.50	0.28

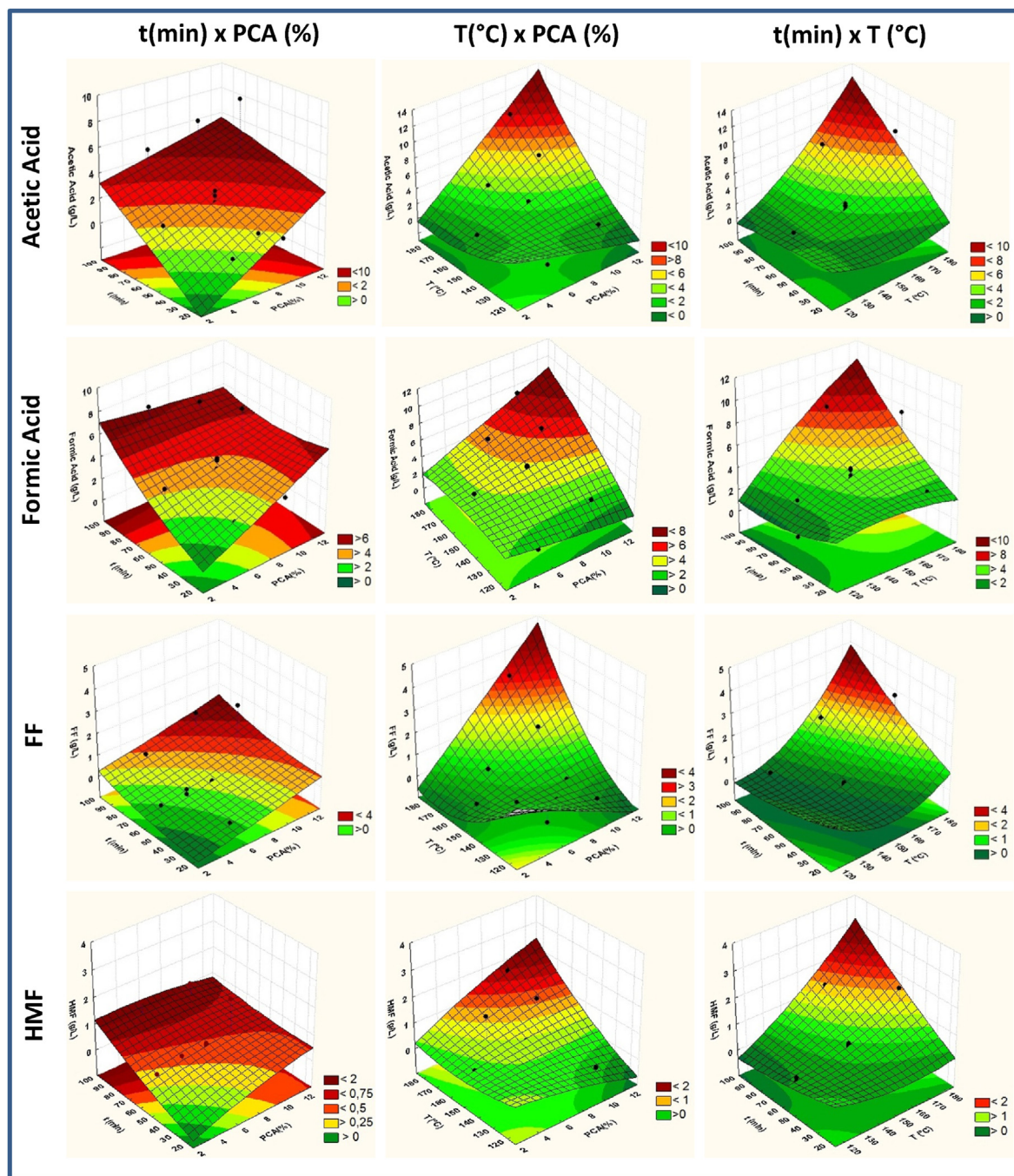


Fig. 1. Response surface graphs generated from the quadratic model, showing the relationship of variables T , t , and PCA with the content of inhibitors (formic acid, acetic acid, HMF, and FF) in the black liquor.

(123.3 kJ mol⁻¹) [44]. In some studies reported in the literature on acid pretreatments, the highest sugar conversion rates were shown to be dependent on reaction time and temperature [45], and that the acid concentration and the extent of hemicelluloses removal and lignin modification can affect enzymatic conversion [15]. In this study, it was found that at long reaction times and high temperatures there was a higher conversion of polysaccharides into sugars.

Analyzing data shown in Table 3 and the response surface graphs shown in Fig. 3, it is possible to observe the pretreatment conditions that led to high *TRS* concentrations after enzymatic hydrolysis of the pretreated SCB. It was observed that for a fixed

reaction time (55 min), the use of higher levels of T and lower levels of PCA or the use of lower levels of T and higher levels of PCA led to higher *TRS* concentrations. In addition, for a fixed reaction temperature (150 °C), the use of higher levels of t and PCA led to higher *TRS* concentrations. Finally, for a fixed PCA (7.50 wt%), the use of intermediate levels of t and T led to higher *TRS* concentrations. It is important to emphasize that arabinose was detected in few samples, and it was not possible to assess the effects of the variables studied on its conversion. However, it is observed that the samples containing arabinose were the samples mostly pretreated at low temperatures.

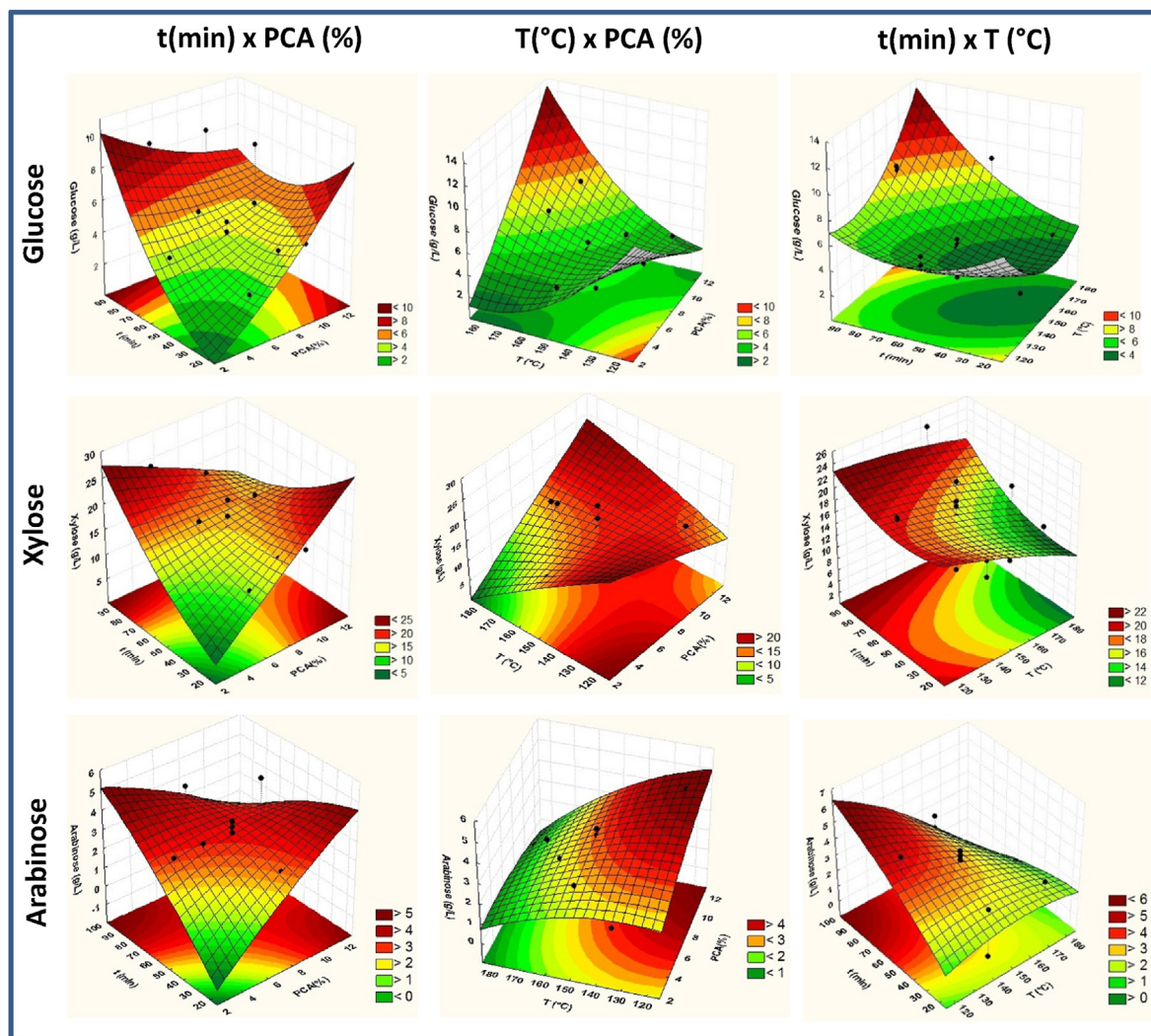


Fig. 2. Response surface graphs generated from the quadratic model, showing the relationship of variables T , t , and PCA with the contents of glucose, xylose, and arabinose in the black liquor.

3.3. Anaerobic biodegradability tests

Pretreatment condition 2, a less severe pretreatment condition in terms of temperature, led to the highest cellulose content (51.72%) in the solid fraction [30] and a high concentration of TRS (171.45 g L^{-1}) after enzymatic hydrolysis. In addition, this condition also led to a high concentration of xylose (20.44 g L^{-1}) and arabinose (5.38 g L^{-1}) in the liquid fraction (BL), with a lower concentration of inhibitors. Therefore, it can be inferred that lower severity pretreatment conditions, even if they imply lower contents of glucose in hydrolysates, should be preferred when the objective is the biogas production. Thus, there is a compromise between cost and feasibility of the proposed process with less waste generation (environmental concern). This way, the fractions resulting from the dilute citric acid pretreatment in condition 2 were chosen for the investigation of biogas production, considering a condition that can be used so that several by-products are produced concomitantly in a biorefinery.

The biogas production depends on COD concentration, i.e., organic compounds contained in pretreated biomass fractions. The BL generated after pretreatment under condition 2 showed a COD of $50.40 \text{ g O}_2 \text{ L}^{-1}$ and the RSF not converted into sugars after enzymatic hydrolysis showed a COD of $1.13 \text{ g O}_2 \text{ L}^{-1}$. The biogas

production results obtained from BBP tests with BL are shown in Fig. 4(A). The behavior of BL in anaerobic digestion showed a significant production in the initial phase. After the 10th day, the highest biogas production was $209.20 \text{ NmL gCOD}^{-1}$, increasing gradually until reaching the maximum average volume of biogas of $563.59(\pm 70.36) \text{ NmL gCOD}^{-1}$ on the 50th day. Due to the absence of the lag phase for anaerobic digestion of BL (initial adaptation phase), the modified Gompertz model was not used. Therefore, it is concluded that the BL obtained from condition 2, with an adequate amount of sugars and a minimal amount of toxic/recalcitrant compounds created a suitable environment for biogas production. Fig. 4(B) shows a graph with average volume of biogas produced (accumulated volume of biogas and accumulated volume of biogas per COD added) by anaerobic digestion of the RSF obtained after enzymatic hydrolysis of pretreated SCB in condition 2, as a function of time.

For RSF (condition 2), biogas production started in the first few hours, and it was promising, increasing gradually up to a maximum volume of biogas of $69.49(\pm 1.87) \text{ NmL gCOD}^{-1}$ on a 50th day. The results regarding biogas production in relation to volatile and total solids of the RSF were $69.42 \text{ NmL gSV}^{-1}$ and $66.38 \text{ NmL gST}^{-1}$, respectively. The RSF that remained in the solid phase after enzymatic

Table 3

Composition of the liquid fractions (hydrolysates) obtained after the enzymatic hydrolysis of samples of raw SCB and pretreated SCB.

Sample	Composition (g L ⁻¹)			
	TRS (g L ⁻¹)/Yield ^a (%)	Glucose ^b	Xylose ^b	Arabinose
SCB	25.86/87.07	12.96	4.28	n.d.
1	10.70/93.91	4.41	1.24	n.d.
2	171.45/83.07	9.86	3.58	0.02
3	68.95/92.46	8.51	2.48	0.03
4	35.19/96.54	6.53	1.97	0.03
5	58.15/94.16	7.20	1.96	0.04
6	99.10/59.09	14.37	2.70	n.d.
7	76.59/77.66	20.27	4.40	n.d.
8	164.73/77.27	18.44	4.48	n.d.
9	184.79/76.91	19.92	5.23	n.d.
10	52.73/81.71	16.06	2.64	n.d.
11	103.99/82.73	16.20	4.31	n.d.
12	29.35/83.06	15.16	2.27	0.06
13	89.08/72.90	18.34	2.67	n.d.
14	36.11/44.31	23.74	3.62	n.d.
15	45.10/90.50	14.80	3.86	n.d.
16	60.13/37.34	21.48	1.60	n.d.
17	74.94/11.23	23.70	0.23	n.d.

n.d. = not detected.

^a Yield (%) = $\left(\frac{\text{dry matter after enzymatic hydrolysis}}{\text{dry matter before enzymatic hydrolysis}} \right) \times 100$

^b Data previously published by our research group [30].

hydrolysis also appears to have become accessible to cellulolytic enzymes during the anaerobic digestion, leading to the gradual release of sugars, thus contributing to biogas production for a longer time [46]. The obtained results showed shorter lag periods (initial adaptation phase), associated with biodegradability and positive effects on the biogas production. In addition, the observed biogas production profile shown in Fig. 4(C) was similar to that predicted by the modified Gompertz model. From a chemical composition standpoint, the higher initial biogas production can probably be related to the conversion of degradable substrates, such as residual fraction of non-hydrolyzed cellulose and lignin-hemicelluloses complex in the previous step (enzymatic hydrolysis) [1]. The analysis of the estimated kinetic parameters [5] combined with the levels of the variables *T*, *t*, and *PCA*, as well as the chemical composition of the fractions used for biogas production confirmed the benefits of implementing anaerobic digestion in the proposed process, which can be used to take advantage of highly recalcitrant residues, such as non-enzymatically hydrolyzed pretreated solid residues, for biogas production.

When evaluating biogas production, it is highlighted that black liquor, composed of sugars such as xylose and arabinose, can be easily fermented. Regarding the non-hydrolyzed solid fraction obtained after enzymatic hydrolysis, there was an increase in the biogas production, but with a lag phase, due to the high content of residual cellulose in the sample. In both cases, the high rates of conversion and the high production of biogas in volume are highlighted, proving the feasibility of the pretreatment method with dilute citric acid and its application to sugarcane bagasse associated with the production of other by-products in addition to bioethanol.

Some studies were carried out by Amnuaycheewa et al. [47], with non-pretreated and pretreated rice straw subjected to anaerobic digestion to evaluate the influence of organic acid pretreatment on biogas production. The highest biogas yield was 322.1 mL per gram of rice straw pretreated with citric acid, which was 7.40 times higher than for non-pretreated biomass. Rattanaporn et al. [48] used three organic acids, including acetic acid, oxalic acid, and citric acid for the pretreatment of oil palm trunk biomass, and investigated the improvement provided by pretreatment in the enzymatic saccharification and bioethanol production. Citric acid

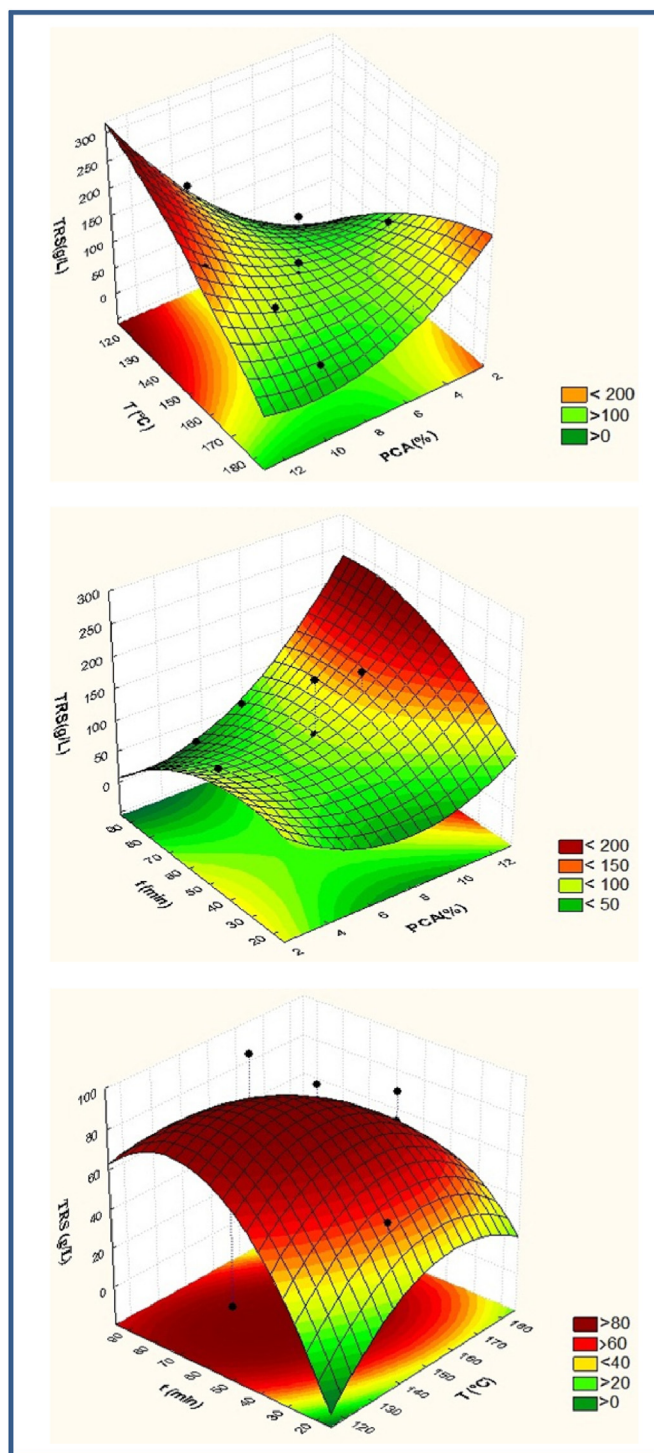


Fig. 3. Response surface graphs generated from the quadratic model, showing the relationship of variables *T*, *t*, and *PCA* with TRS concentration in the liquid fraction (hydrolysates) obtained after the enzymatic hydrolysis of pretreated SCB samples.

pretreatment was the most efficient pretreatment method to improve the fermentation of sugars present in the hydrolysate with *Saccharomyces cerevisiae* TISTR 5606 to bioethanol, with a yield 1.94 times higher than that for non-pretreated biomass. These results demonstrated that organic acids can be used for the pretreatment of lignocellulosic materials as a potential method to increase enzymatic hydrolysis yield and consequently biogas/biofuel production.

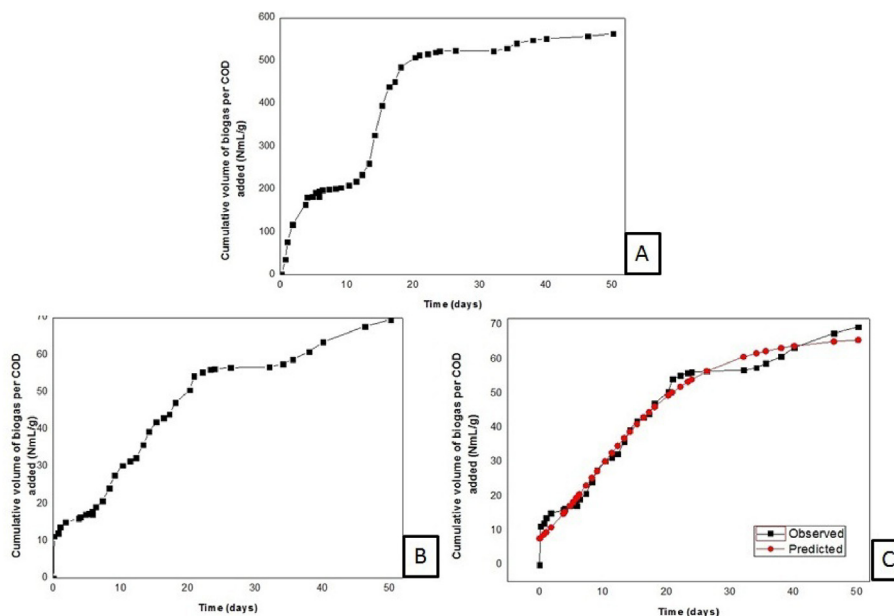


Fig. 4. Biogas production using (A) liquid fraction (black liquor) and (B) residual solid fraction (RSF) obtained from enzymatic hydrolysis of the pretreated solid fraction (condition 2) and (C) biogas production using RSF obtained after enzymatic hydrolysis (condition 2) predicted by the modified Gompertz model, with the accumulated volume of biogas produced per COD added, as a function of time.

Therefore, the present study allowed to validate the use of a dilute citric acid pretreatment of sugarcane bagasse and also opens the way for further studies involving the production of other by-products in addition to bioethanol, since there are no previous studies of pretreatment of SCB with dilute citric acid for biogas production. In addition, the information collected in the present study will be useful to guide new experimental conditions, seeking new strategies for an economic and feasible process (due to the consumption of energy and inputs), as well as a process of greater productivity with greater profitability, but environmentally safer.

3.4. Estimation of the economic-energy feasibility of the proposed process

As previously discussed, the black liquor (BL) obtained from dilute citric acid pretreatment of SCB in condition 2 and the residual solid fraction (RSF) obtained after enzymatic hydrolysis of pretreated SCB in condition 2 were subjected to anaerobic digestion tests to evaluate the BBP of these fractions. A brief energy and economic assessment were carried out considering the energy balance for the dilute citric acid pretreatment, i.e., thermal energy

generated by burning biogas against thermal energy spent in the pretreatment, as well as the revenues from a possible sale of the electricity generated. Therefore, two scenarios were considered: i) ambient water temperature ($T_i = 25\text{ }^{\circ}\text{C}$) and ii) process water temperature, i.e., in the vapor state ($T_i = 100\text{ }^{\circ}\text{C}$). The results obtained are presented in Table 4.

As previously reported by Ciriminna et al. [19] and Gomes et al. [30], the pretreatment with dilute citric acid can improve the biodegradability of sugarcane bagasse with low generation of inhibitors. As seen in Table 2, condition 2 allowed the solubilization of a high amount of xylose and arabinose in the liquid fraction (BL), with a low generation of inhibitors (HMF and FF). This certainly corroborated the high methane production ($563.59\text{ NmL gCOD}^{-1}$) from BL, which resulted in $2776\text{ MJ ton}_{\text{SCB}}^{-1}$ of energy.

Pretreatment of biomass can generate various compounds that act as inhibitors (i.e., acetic acid, levulinic acid, FF, HMF, and phenolics), which negatively impact cell viability and productivity, and subsequently affect bioethanol and biogas production [47]. These inhibitors cause the membrane leakage, which leads to cell death and failure during the fermentation process. Furan derivatives, such as FF and HMF, generated from the degradation of sugars and small

Table 4
Estimation of the economic-energy feasibility of the proposed process.

Proposed scenarios	$ED\text{ (MJ ton}_{\text{SCB}}^{-1})^a$	$TEP\text{ by AD (MJ ton}_{\text{SCB}}^{-1})^b$	$AE\text{ (MJ ton}_{\text{SCB}}^{-1})^c$	$EE\text{ (kWh ton}_{\text{SCB}}^{-1})^d$	$TE\text{ (MJ ton}_{\text{SCB}}^{-1})^e$	$EEP\text{ (USD ton}_{\text{SCB}}^{-1})^f$	$TEB\text{ (MJ ton}_{\text{SCB}}^{-1})^g$	$TP\text{ (USD ton}_{\text{SCB}}^{-1})^h$
1 – $T_i = 25\text{ }^{\circ}\text{C}$								
BL-AD	1990.25	2776.26	2359.82	229.43	1533.88	18.35	- 456.37	18.35
RSF-AD	1990.25	1.66	1.41	0.14	0.92	0.01	- 1989.33	0.01
2 – $T_i = 100\text{ }^{\circ}\text{C}$								
BL-AD	419.00	2776.26	2359.82	229.43	1533.88	18.35	1114.88	18.35
RSF-AD	419.00	1.66	1.41	0.14	0.92	0.01	- 418.08	0.01

^a ED: Thermal energy demand of dilute citric acid pretreatment for 1 ton of sugarcane bagasse.

^b TEP by AD: Total energy production per ton of sugarcane bagasse via anaerobic digestion.

^c AE: Available energy considering a CHP (combined heat and power) system.

^d EE: Electric energy considering the conversion of available energy of 35% ($1\text{ MJ} = 0.277778\text{ kWh}$).

^e TE: Thermal energy considering the conversion of available energy of 65% ($1\text{ MJ} = 0.277778\text{ kWh}$).

^f EEP: Electrical energy profitability: Profitability of the sale of electricity generated ($\text{USD } 0.08\text{ kWh}^{-1}$ [40]).

^g TEB: Thermal energy balance (thermal energy generated ($E^{(+)}$) - thermal energy consumed ($E^{(-)}$).

^h TP: Total profitability.

organic acids obtained as by-products from degradation of hemicelluloses have also been found to stop the growth of fermenting microbes, depleting the supply of nicotinamide adenine dinucleotide phosphate (NADPH), which is necessary for energy biosynthesis and therefore causing membrane leakage. In addition, small organic acids reduce microbial growth by interfering with cytoplasmic pH homeostasis [48].

As shown in Table 4, the first scenario that couples dilute citric acid pretreatment of SCB with anaerobic digestion is only energetically viable if the anaerobic digestion of the BL produced is performed. Even so, the thermal energy balance will only be positive if, and only if, the initial temperature of the water used in the pretreatment is higher than 47 °C. In this scenario, thermal energy consumed is equal to thermal energy generated. In the second scenario proposed by this study, the resulting thermal energy supplied the energy demanded, with a surplus of almost 1115 MJ ton⁻¹SCB, which can be used to heat the inlet water used in the pretreatment itself.

The fact that only condition 2 with anaerobic digestion of BL (BL-AD) had a positive thermal energy balance is justified by the lower heating interval (20 °C), since dilute citric acid pretreatment has the energy consumption as a limiting factor, as it requires high amount of energy to heat the water to the set reaction temperature. Therefore, the longer the water heating interval, the more difficult it will be to supply the energy demand of the proposed process. In addition, under these conditions, the electrical energy generated by the proposed process (~229 kWh ton⁻¹SCB) can be sold to local distribution concessionaires or used in the processing plant, contributing to an increased revenue.

4. Conclusions

The influence of parameters such as PCA, *T*, and *t* in the dilute citric acid pretreatment of sugarcane bagasse was investigated, with biogas production from black liquor (BL) and residual solid fraction (RSF) obtained after enzymatic hydrolysis. The pretreatment carried out under milder *T*, *t*, and PCA conditions also resulted in a more accessible and cellulose-rich material for cellulolytic enzymes in enzymatic hydrolysis. From the anaerobic digestion (AD) of BL and RSF obtained from the pretreatment in condition 2 (120 °C, 9.75 wt% PCA, 55 min), the biogas production was 563.59 NmL gCOD⁻¹ and 57.78 NmL gCOD⁻¹, respectively. The estimation of the economic-energy feasibility showed that the proposed process has the potential to be implemented, but it needs full energy integration with biogas production by AD of BL.

CRediT authorship contribution statement

Michelle Garcia Gomes: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Aline Gomes de Oliveira Paranhos:** Formal analysis, Writing – original draft, Writing – review & editing. **Adonai Bruneli Camargos:** Formal analysis, Writing – original draft, Writing – review & editing. **Bruno Eduardo Lobo Baêta:** Conceptualization, Methodology, Formal analysis, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Milla Alves Baffi:** Conceptualization, Methodology, Formal analysis, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Leandro Vinicius Alves Gurgel:** Conceptualization, Methodology, Formal analysis, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Daniel Pasquini:** Conceptualization, Methodology, Formal analysis, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was financed in part by the Brazilian Coordination for the Improvement of Higher Education Personnel (CAPES) – Finance Code 001, and the Minas Gerais Research Foundation (FAPEMIG) – grant numbers APQ-01360-14 and PPM-00645-17. The authors are also grateful for the financial support from the Brazilian National Council for Scientific and Technological Development (CNPq) and the Brazilian Study and Project Funding Agency (FINEP).

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